

Beverage can proportional counter

Eetu Heikkila, Jeff Schymiczek
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Abstract

A proportional counter particle detector was constructed from an aluminium beverage can through which a 50 μm copper wire was threaded as multiplication anode. By measuring the multiplication factor at bias voltages between 900 and 2200 V, Diethorn's formula for the multiplication factor of a cylindrical proportional counter was confirmed. Afterwards, the detector was used to perform X-ray spectrometry on ^{55}Fe , ^{241}Am and ^{133}Ba . We were able to recover the most distinct peaks below 60 keV cited in literature with good resolution.

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1 Introduction

The work in this lab report is based on an experimental setup published by our supervisors in Winkler et. al. [8] as well as instructional material presented during the course lectures. The detector we built operates on the proportional counter design, using avalanche electron multiplication. Here, ionising gamma radiation create ion-electron pairs. Electrons created in this step are then multiplied by the steep gradient of the electric field around the anode wire.



Figure 1: The final built beverage can detector.

A regular 0.5 L aluminium can is used as the cathode for a proportional counter detector by threading an anode wire through it, which acts as avalanche anode. The wire is connected to a nylon screw as well a high voltage connector. The latter is glued into an acrylic window which replaces the can lid. The detector is further fitted to allow for gas flow through it. An accurate formula for the rate of electron multiplication was first derived by Diethorn in 1956 [4]. The first part of the experimental analysis of the detector consists of measuring the collected charge at various bias voltages to confirm Diethorn's formula. By observing the resolution of the detector at various bias voltages, an optimal operating region is determined. The detector is then supplied with this optimal bias voltage and used to take spectra of various X-ray sources (^{55}Fe , ^{241}Am and ^{133}Ba). Clearly recognisable peaks in the measured spectra are identified using literature.

2 Theory

A proportional counter consists of a gas chamber whose walls form a cylindrical capacitor [6]. Charged particles create ion-electron pairs which travel to the anode/cathode and create a current. Since the current generated by primary charge carriers is relatively small, some process needs to be introduced in order to create larger accumulated charges from singular electron-ion pairs. The standard way to do this is to make the electric field inhomogeneous close to the anode. By concentrating the voltage gradient to a small region around the anode, the electrons are accelerated heavily and undergo gas multiplication, i.e. they themselves create electron-ion pairs, whose electrons can then also be accelerated and create even more electrons, and so on. The electric field of a cylindrical capacitor is optimal for this, since it

is anti-proportional to the distance from the centre of the cylinder r :

$$\mathcal{E} = \frac{V}{r \ln \frac{R_C}{R_A}} \quad (1)$$

where R_C is the cathode radius and R_A is the anode radius.

In order to model the process of gas multiplication, let σ be the (field strength dependent) cross-section of a secondary ionisation. The probability of such an ionisation is given by the Townsend coefficient $\alpha = \sigma N$, where N is the number density of the gas particles. Since the change in the number of electrons n is related to the number of electrons by this coefficient, we have:

$$n(x) = n_0 e^{\int^x \alpha(x') dx'} \quad (2)$$

We define the multiplication factor as $M = \frac{n_A}{n_0}$, where n_A is the number of electrons hitting the anode. Assuming that the electrons require a minimum energy to ionise further atoms, we can imagine that due to equation (8), the region where charge multiplication can occur $r < r_{\text{critical}}$, is small compared to the volume of the cylinder. Since $\alpha \approx 0$ in this region, we have

$$\ln M = \int_{R_A}^{r_{\text{critical}}} \alpha(r) dr \quad (3)$$

Diethorn [4] assumed that depending on gas pressure P , there is a minimal electric field strength KP for which charge multiplication can occur. With this assumption we have:

$$\ln M = \int_{KP}^{\mathcal{E}(R_A)} \alpha(\mathcal{E}) \frac{dr}{d\mathcal{E}} d\mathcal{E} = \frac{V}{\ln \frac{R_C}{R_A}} \int_{KP}^{\mathcal{E}(R_A)} \frac{\alpha(\mathcal{E})}{\mathcal{E}^2} d\mathcal{E} \quad (4)$$

He further assumed that an electron needs to travel through a mean voltage difference ΔV in order for ionisation to occur and that from each point of ionisation two electrons branch out that can engage in charge multiplication. Thus $\alpha = \frac{\ln 2}{\Delta V} \mathcal{E}$ and

$$\ln M = \frac{V \ln 2}{\Delta V \ln \frac{R_C}{R_A}} \int_{KP}^{\frac{V}{R_A \ln \frac{R_C}{R_A}}} \frac{1}{\mathcal{E}} d\mathcal{E} = \frac{V \ln 2}{\Delta V \ln \frac{R_C}{R_A}} \left(\ln \left[\frac{V}{R_A \ln \frac{R_C}{R_A}} \right] - \ln(KP) \right) \quad (5)$$

$$= \frac{V \ln 2}{\Delta V \ln \frac{R_C}{R_A}} \ln \left(\frac{V}{K P R_A \ln \frac{R_C}{R_A}} \right), \quad (6)$$

where ΔV and K are experimentally determined material constants with considerable uncertainties. We have for error propagation:

$$\begin{aligned} \sigma_{\ln M} &= \sqrt{\left(\frac{\partial \ln M}{\partial \Delta V} \sigma_{\Delta V} \right)^2 + \left(\frac{\partial \ln M}{\partial K} \sigma_K \right)^2 + \left(\frac{\partial \ln M}{\partial P} \sigma_P \right)^2 + \left(\frac{\partial \ln M}{\partial R_A} \sigma_{R_A} \right)^2 + \left(\frac{\partial \ln M}{\partial \ln \frac{R_C}{R_A}} \sigma_{\ln \frac{R_C}{R_A}} \right)^2} \\ &= \sqrt{(\ln M)^2 \left[\left(\frac{\sigma_{\Delta V}}{\Delta V} \right)^2 + \left(\frac{\sigma_{\ln \frac{R_C}{R_A}}}{\ln \frac{R_C}{R_A}} \right)^2 \right] + \left(\frac{V \ln 2}{\Delta V \ln \frac{R_C}{R_A}} \right)^2 \left[\left(\frac{\sigma_K}{K} \right)^2 + \left(\frac{\sigma_P}{P} \right)^2 + \left(\frac{\sigma_{R_A}}{R_A} \right)^2 + \left(\frac{\sigma_{\ln \frac{R_C}{R_A}}}{\ln \frac{R_C}{R_A}} \right)^2 \right]} \end{aligned} \quad (7)$$

For the P-10 gas we used, literature values are quoted as $\Delta V = 23.6 \pm 5.4$ V and $K = (4.8 \pm 0.3) \times 10^4 \text{Vcm}^{-1} \text{atm}^{-1}$ in [9].

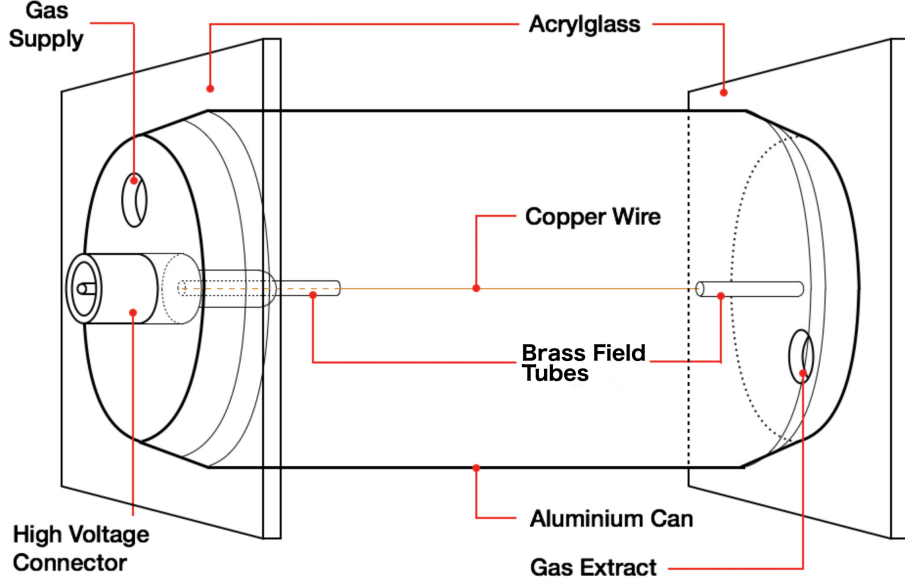


Figure 2: Schematic drawing of proportional counter (Source: instruction material, originally by Joshua Read)

The collected charge Q can be experimentally determined by assuming that the decay time of the capacitor in the pre-amplifier is a lot longer than the rise time of the detector signal. Under this approximation, the charge is simply $Q = CV$. The multiplication factor can be then calculated simply as

$$M = \frac{Q}{n_0 e} = \frac{Q}{e \frac{E_{\text{incident}}}{E_{\text{ionpair}}}} = \frac{C V E_{\text{ionpair}}}{e E_{\text{incident}}}, \quad (8)$$

where E_{incident} is the energy of the incident radiation and $E_{\text{ionpair}} = 26 \text{ eV}$ [9] is the energy required to create one electron-ion pair.

We have for the standard deviation of the collected charge $\frac{\sigma_Q}{Q} = \sqrt{\frac{1}{n_0}(F + b)}$, where $F = 0.17$ [6] is the Fano factor correcting for the smaller-than-Poisson uncertainty of the number of ionised particles and $b = 0.5$ [6] is a factor stemming from the Polya distribution followed by the number of electrons created in the gas multiplication avalanche. We get a theoretical minimal resolution of:

$$r = \frac{\text{FWHM}}{E} = 2.35 \frac{\sigma_Q}{Q} = 2.35 \sqrt{\frac{E_{\text{ionpair}}}{E_{\text{incident}}}(F + b)} = 9.8 \sqrt{\text{eV}} \frac{1}{\sqrt{E_{\text{incident}}}} \quad (9)$$

Thus we would have a minimum of 12.8% resolution for our measurement of ^{55}Fe and 4.0% for ^{241}Am .

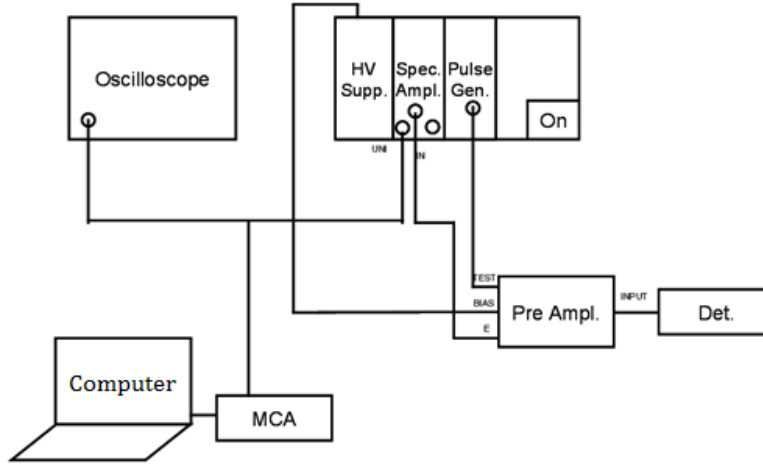


Figure 3: Schematic drawing of the peripheral electronics (Source: instruction material)

3 Experimental Setup

The proportional counter was constructed out of a 0.5 L beverage can of diameter $d_C = 65.5 \pm 0.5$ mm and height $h = 160 \pm 0.5$ mm. Along the central axis of this can, a copper anode wire of diameter $d_A = 50 \pm 2 \mu\text{m}$ was threaded. A schematic view of the detector can be seen in figure 2. Brass field tubes of diameter $d_B = 1.52 \pm 0.01$ mm shield the wire in the front and back of the can and prevent gas multiplication to happen in places where the electric field is inhomogeneous. The back brass tube is fixed using friction into a nylon screw. This screw is mechanically tightened into a hole in the back of the can. The anode wire is fixed to the flat end of the screw by soldering it onto the brass tube. The lid of the can is completely removed and replaced by an acrylic window, in the centre of which a high voltage connector (HV) has been glued. Before this assembly step, the plastic lining on the inside of the can has been removed using an abrasive brush. A brass tube as well as the anode wire are soldered onto the HV-connector. A similar acrylic window is glued on the back end of the can to ensure a gas-tight seal around the can as well as to act as a mechanical support to level the can. Holes are drilled into both acrylic windows and the back of the can to allow for gas exchange. The holes in the acrylic windows are fitted with gas tubes. All parts of the detector have been cleaned in an ultrasound bath before assembly.

The can is filled with P-10 gas (90% Argon, 10% Methane). Due to its inertness, relatively high atomic number and affordability, Argon is a suitable detector gas for a proportional counter. The methane acts as a quenching gas, which will absorb spurious photon emissions from excited atoms.

In order to determine the effective gas volume of the detector we measured the extent of the brass field tubes into the can volume. The tubes extended along the nylon screw from the back end of the can by $d_1 = (22.15 \pm 0.05)$ mm. From the front end we measured $d_2 = (32 \pm 1)$ mm. This gives us a total effective length of nylon wire of $l = h - d_1 - d_2 = 105.8 \pm 1$ mm

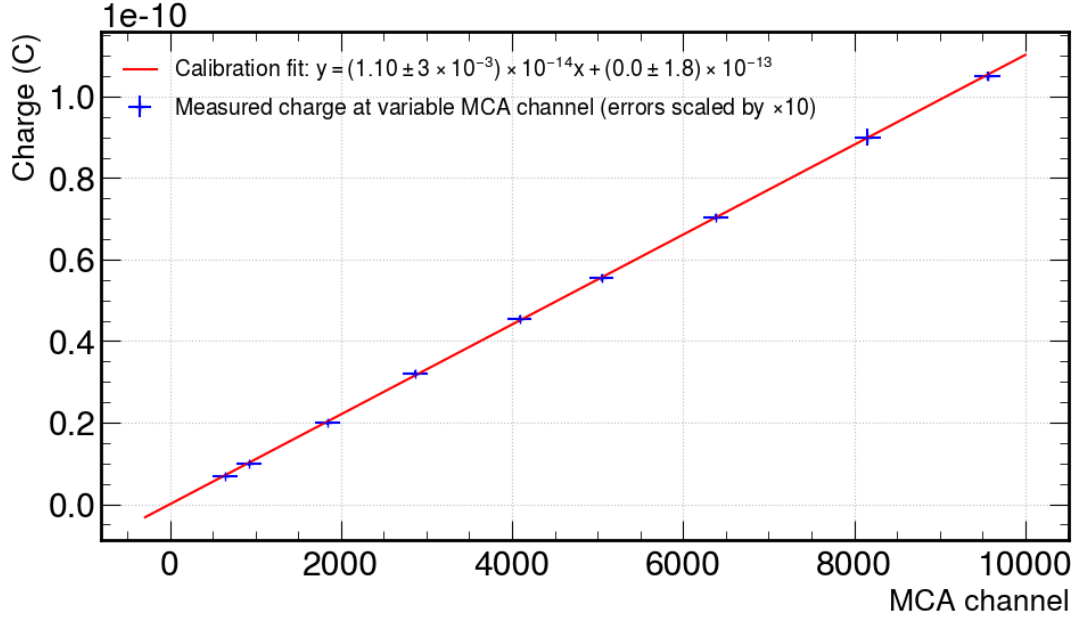


Figure 4: Electronics calibration curve produced by feeding a signal with known voltage to the detector using a tail pulse generator. A Gaussian is fit to the measured output signal to obtain the centroid MCA channel corresponding to an input voltage. Resultant charge is inferred using the $Q = CV$ approximation with $C = 1$ pF and V the input voltage. Measurement errors in the plot are scaled up by a factor of 10 for visibility.

and an effective gas volume of:

$$V_{\text{eff}} = \frac{\pi}{4}l(d_C^2 - d_A^2) \approx \frac{\pi}{4}ld_C^2 = (0.357 \pm 0.006)\text{L}$$

Using $\mathcal{E}_{\text{min}} = KP = 48000 \text{ V cm}^{-1}$ as the minimum field for charge multiplication and equation (8), we have that $V > 25000\text{V}$ in order for gas multiplication to occur at the field tube and $V < 900\text{V}$ in order for no gas multiplication to occur at the anode wire. The first constraint has almost no effect on our experiment, while we will see that the second constraint matches very well with the smallest voltage at which we observed a measurable current pulse. The peripheral devices the detector was connected to can be seen in figure 3. The high voltage supply, spectral amplifier and pulse generator were powered by a Plein & Baus UEP15 power supply. The detector bias voltage was supplied by an iseg NHQ 206L high voltage DC power supply. The signal from the detector was first fed into an ORTEC model 142A preamplifier, then into an ORTEC 855 dual spectral amplifier with pulse shaping, whose coarse gain was adjusted such that the Amptek MCA8000D multichannel analyser operating in a 0-1V range would capture the signal. The output of the pulse shaper is also fed into an oscilloscope for testing and debugging purposes. The preamplifier uses a $C = 1$ pF feedback capacitor across the internal operational amplifier on which the incoming current is integrated [1].

The detector was calibrated using a Berkeley Nucleonics Corp BH-1 tail pulse generator at a spectral amplifier course gain of 10. Using this pulse generator we could create pulses

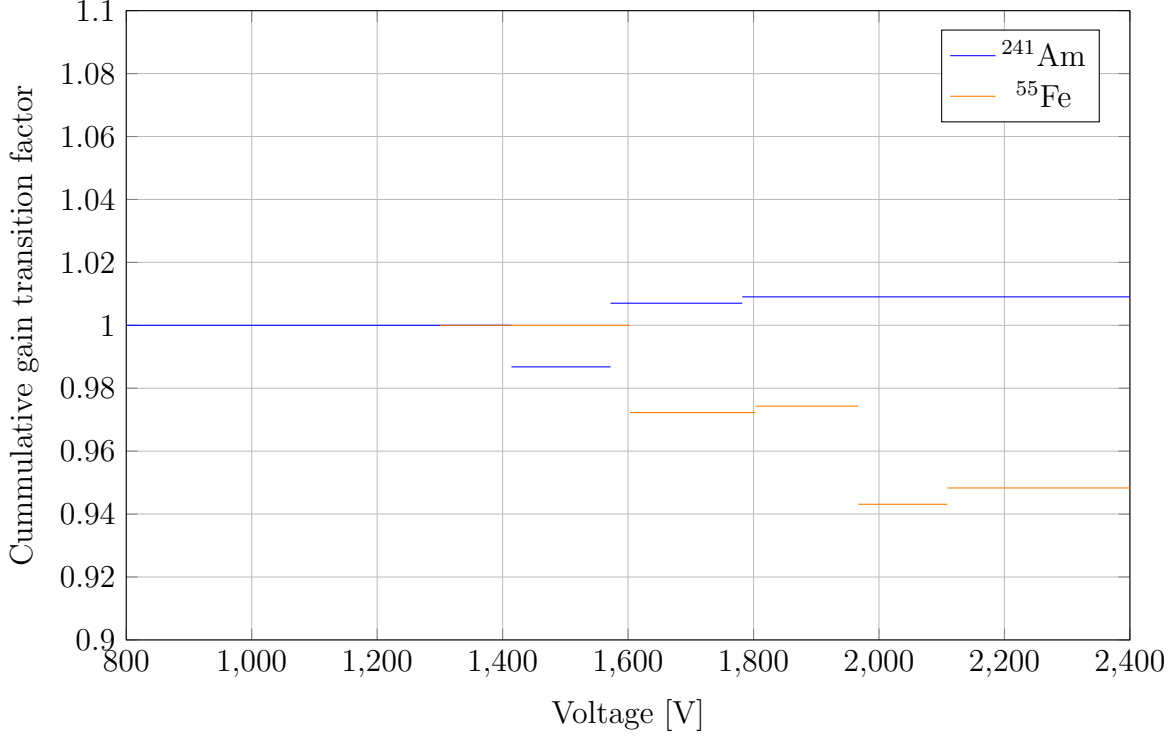


Figure 5: Gain transition curve observed during the two measurement runs.

similar to those produced by the detector. Assuming that the decay time of the collected charge in the preamplifier is a shorter than the pulse time, we could use the simple relationship $Q = VC$ to transfer a voltage read off on the oscilloscope to a collected charge. We then also record the channel where the pulse was registered on the MCA and obtain a calibration graph (figure 4). We observe that the relationship between MCA channel and collected charge is almost proportional with

$$Q = (1.10 \pm 0.03) \times (10^{-14}\text{C})N_{\text{MCA}} + 0.0 \pm 1.8 \times 10^{-13}\text{C}. \quad (10)$$

For the multiplication factor determination, we swept the bias voltage from the minimal voltage at which we could observe a signal peak at the highest gain (100) to the voltage where the signal peak moves past a 1V spectral amplifier signal at minimal gain (2) in roughly 50 V steps. We did this for the distinct 5.9 keV ⁵⁵Fe and 59.5 keV ²⁴¹Am peaks. While doing so, we were able to observe a gain transition graph as in figure 5, by calibrating on the centroid position of the observed peak. The observed gain remained within 5% of the set value of the gain. The gain shift can be explained both by nonlinearities in the electronics and errors fitting to peaks at lower bin amounts. Since we could not separate the two effects, we do not factor gain transitions further into our analysis.

After confirming that the detector's multiplication factor follows equation 6 and finding a suitable voltage for optimal resolution, spectra of background radiation, Ba-133, Am-241 and Fe-55 were taken over at least 10 minute intervals using both collimated and uncollimated sources. Sources were collimated by placing a lead shield with a slid between the source and the detector.

4 Analysis and results

4.1 Confirming Diethorn's Equation

In order to characterize the detector we performed measurements of the easily identifiable ^{55}Fe and ^{241}Am peaks. These were the singular 59.5 keV Am-241 peak and the $5.86 + 6.49$ keV combined Fe-55 peaks. Taking into account the known relative intensities, the combined Fe-55 peak should occur at 5.96 keV. We performed measurements of these peaks at variable voltage and coarse gain, covering the total region where we could identify our chosen peaks. The 59.5 keV Am-241 peak was fit using a Gaussian on a linear continuum. The combined ^{55}Fe photopeaks were fit using a single Gaussian. Fitting was carried out using InterSpec (ref. [2]). We are able to obtain values for the centroids and FWHM in terms of the MCA channel for each of the measurements.

We now can use the measured centroids to confirm that the constructed detector adheres to the theoretical formulation for the multiplication factor of a cylindrical proportional counter according to Diethorn (ref. [4]), stated in equation 6. Using the experimentally obtained charge-voltage calibration fit (equation 10), we can use equation 8 to calculate the charge multiplication at a given bias voltage. Plotting the measured charge multiplication for peaks at various bias voltages we obtain the orange (^{55}Fe) and blue (^{241}Am) series in figure 6. The theoretical prediction is shown in black with the error calculated using equation 7 shown in red. We observe a good fit of the measured multiplication factor to the theoretical prediction by Diethorn's formula.

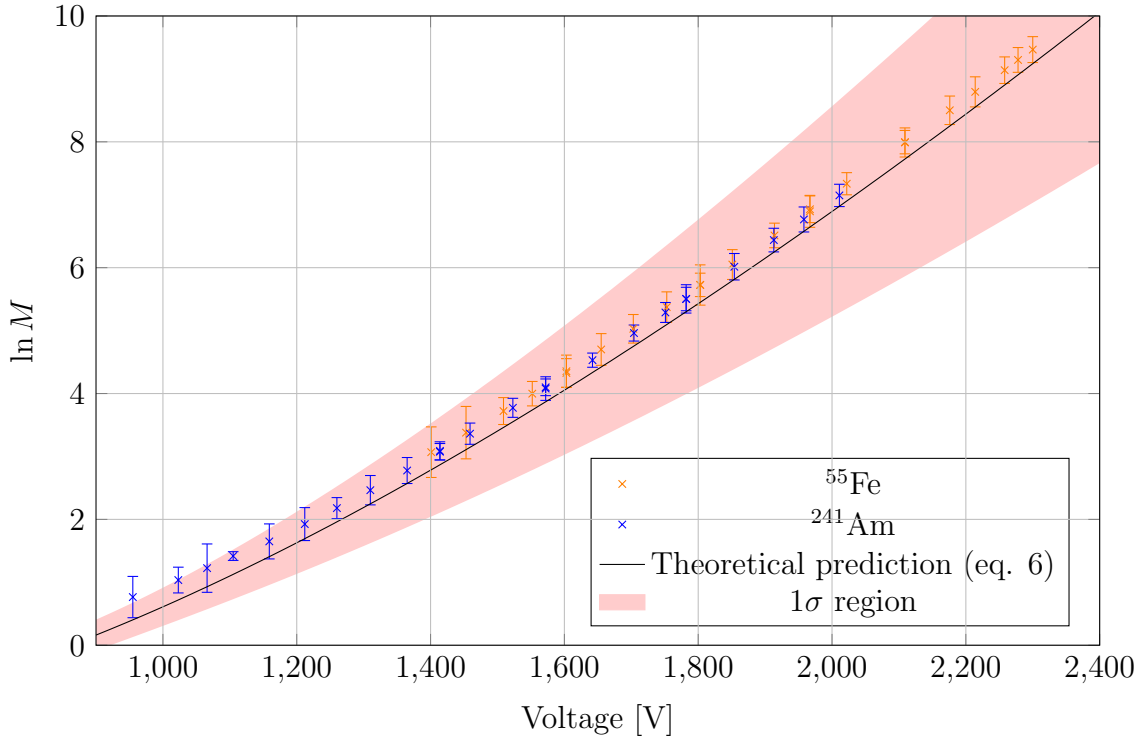


Figure 6: Multiplication Factor. The large error on the theoretical prediction mainly stems from errors in the ΔV factor of equation 6

4.2 Determining detector's optimal working regime

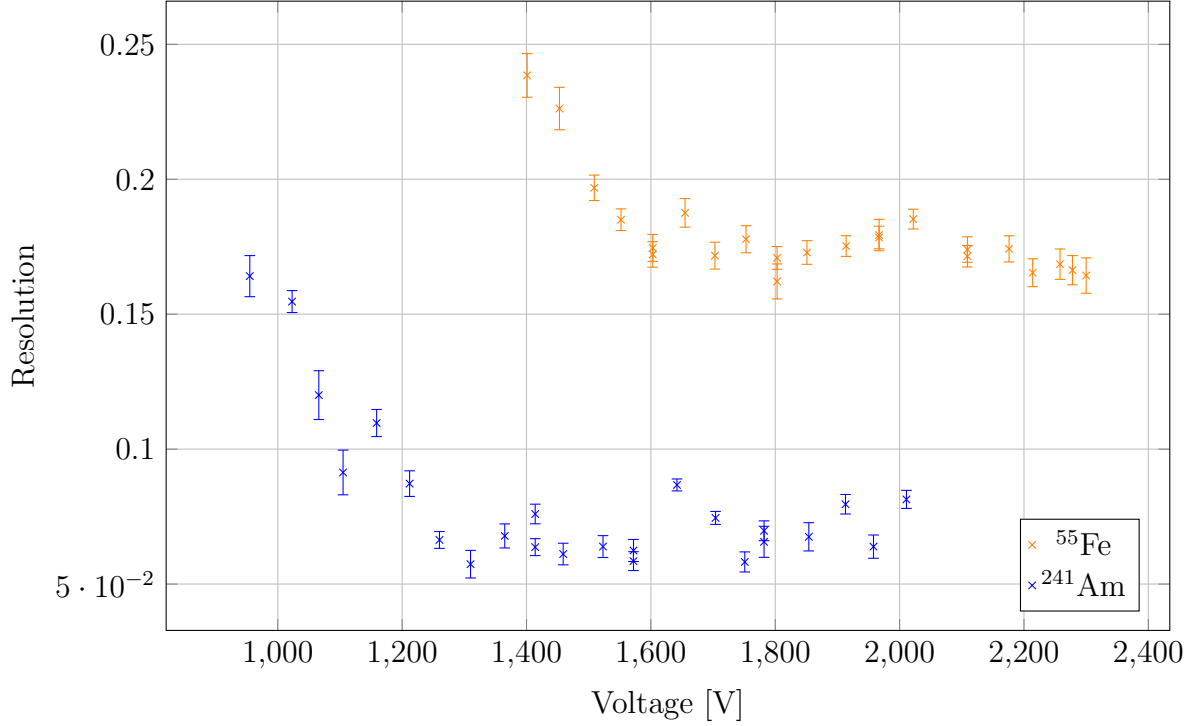


Figure 7: Measured values for the resolution determined as $\frac{N_{MCA}}{FWHM_{MCA}}$.

The resolution calculated as $\frac{N_{MCA}}{FWHM_{MCA}}$ is shown in orange (^{55}Fe) and blue (^{241}Am) in figure 7. We observe a coinciding minimum in the resolution around 1700 - 1900 V which seems to be followed by a plateau for the resolution of both radiation sources. For lower voltages we have lower charge multiplication and poorer charge collection. As we increase the bias voltage, we reach a relatively stable resolution for both sources. For higher bias voltages, we expect deviations from the plateau toward worse resolution due to secondary effects which may degrade the signal. It is unclear whether we already enter this regime in the above figure. Therefore the coinciding minimum is the optimal working bias voltage of the detector.

In the following analysis, the spectral measurements were performed at 1747 V with a coarse gain of 10, which is located within the optimal working regime. Using these settings, the 59.5 keV ^{241}Am peak was visible on the detector with enough space to the right to potentially see the 80 keV ^{133}Ba peak, while the ^{55}Fe peak could still be resolved.

4.3 Energy calibration

The raw measured spectra are in MCA channel vs. measured counts histogram format. In order to transform the raw measurements into energy spectra we needed to perform an energy calibration. Energy calibration was performed using the full spectral measurements performed at a coarse gain of 10 and an applied voltage of 1747 V by fitting a linear line between the measured MCA channel number of the centroid for each of the known peaks, and their known energy. The background radiation was taken into account during the energy calibration phase. The live time for the background radiation measurement was

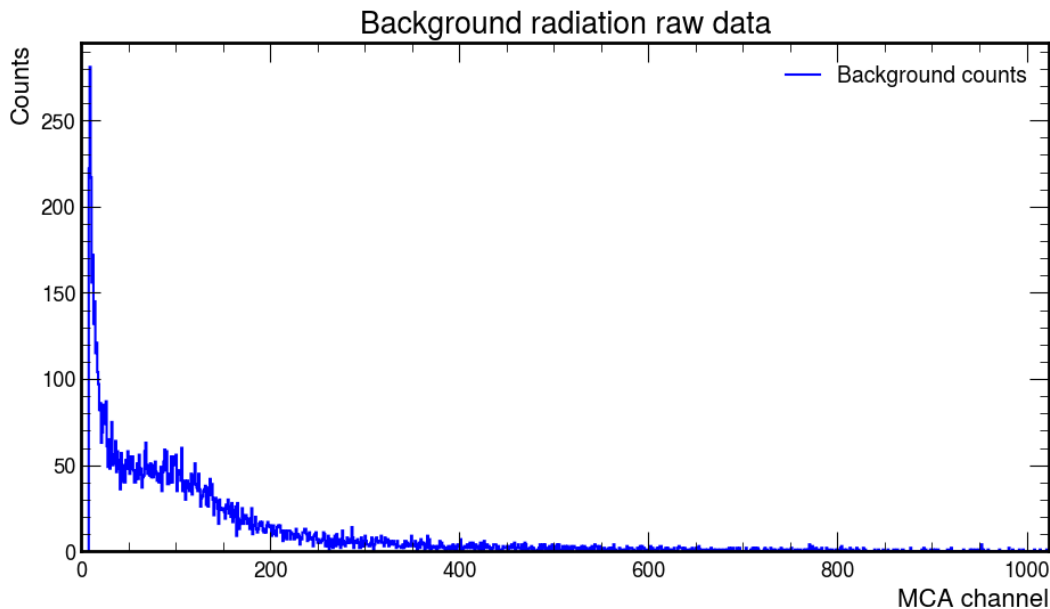


Figure 8: Raw measurement of the background radiation over a period of ~ 30 minutes.

1850 seconds. The background radiation counts were normalized to the live time of the spectra and subtracted from the counts. The raw spectrum for the background radiation is shown in figure 8.

The known peaks used for the energy calibration are the ^{55}Fe combined $K\alpha - 1, 2$ X-ray peaks which should be resolved at an energy of 5.8592 keV, the ^{55}Fe $K\beta$ X-ray peak which should occur at 6.49045 keV, and the ^{241}Am 59.5409 keV γ -ray primary photopeak which is formed as a byproduct of the isotope's main α -particle decay channel [7]. Due to poor resolution at very high energies, the ^{241}Am peak has poor statistics compared to the iron peaks, but was still measurable. Figures 9, and 10 show the fits made in order to determine the MCA channel of the known peaks using the given settings in the optimal regime. The ^{55}Fe peaks were fit as a simple double Gaussian, while the ^{241}Am peak is fit as a Gaussian on a linear continuum. The resulting energy calibration fit is shown in figure 11.

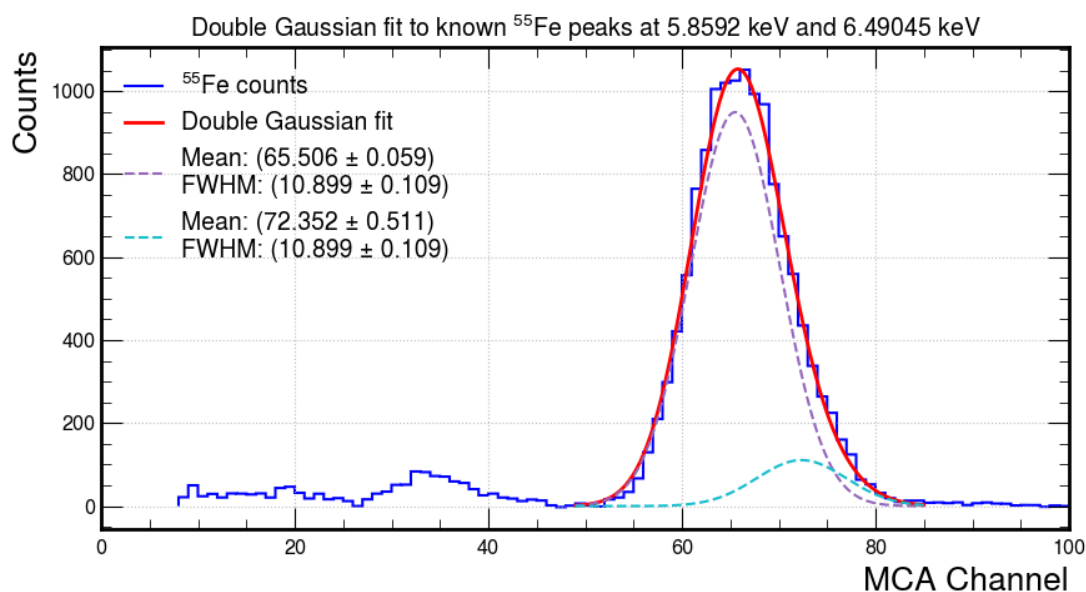


Figure 9: A double Gaussian fit made to the known ^{55}Fe peaks at 5.8592 keV and 6.49045 keV. The background has been subtracted. The full width-half maxima were forced equal for the two peaks because positive solutions were not found by the fitting routine otherwise.

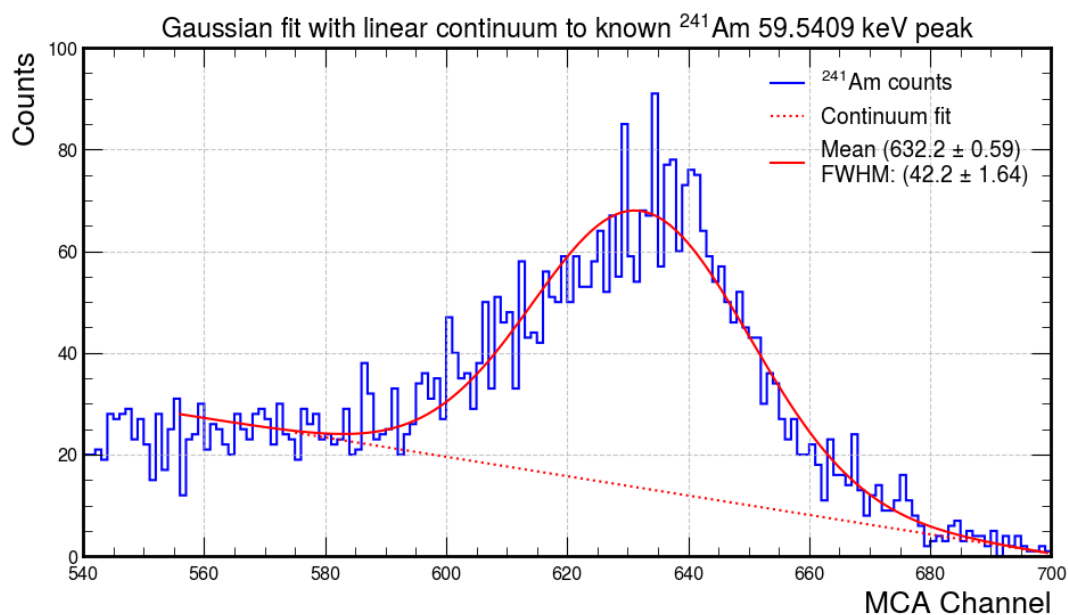


Figure 10: A Gaussian with a linear continuum fit to the known 59.5409 keV ^{241}Am peak.

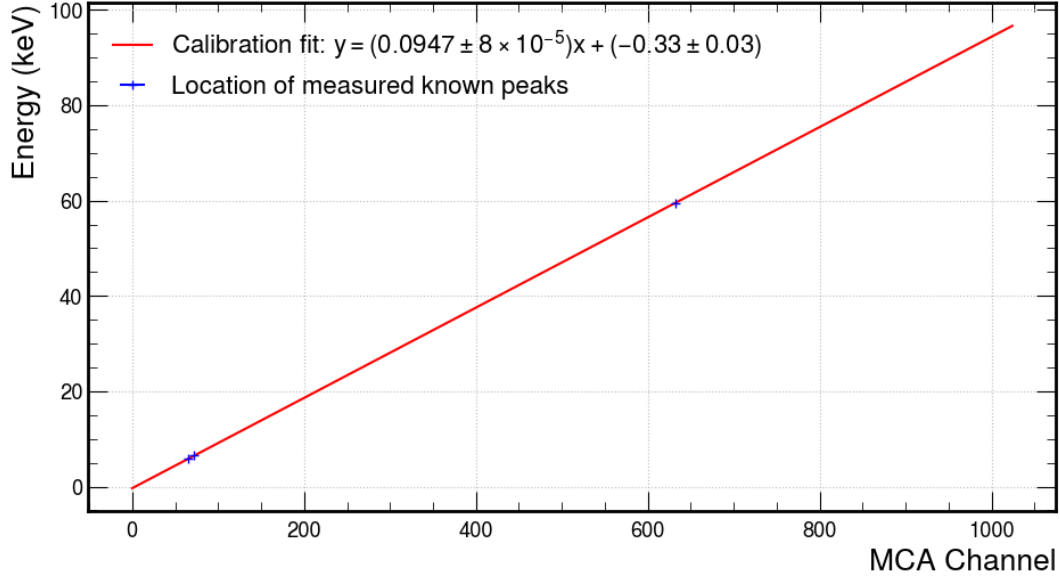


Figure 11: An energy calibration fit produced by fitting for the MCA channel of known peaks in the full spectral measurements. The fit is made to the plot of measured MCA channel vs. corresponding known peak energy.

The energy calibration fit can be used to determine the energy E in keV of a measurement corresponding to an MCA channel N_{MCA} using the linear relationship

$$E = (0.0947 \pm 8 \times 10^{-5})N_{\text{MCA}} + (-0.33 \pm 0.03) \quad (11)$$

There is an inherent systematic error in the fit, because it does not perfectly pass through $(N_{\text{MCA}}, E) = (0, 0)$ on the axis. We should keep this error in mind when identifying unknown peaks in the following discussion. When judging the fit in figure 10 by eye it could be argued that the peak should be located ~ 5 MCA channels to the left. This is likely an artifact of choosing too large a region of interest for the fit. Because this point is at a much greater energy compared to the ^{55}Fe peaks, it has a high leverage in the linear fit. This may be the reason for the imperfect vertical intercept.

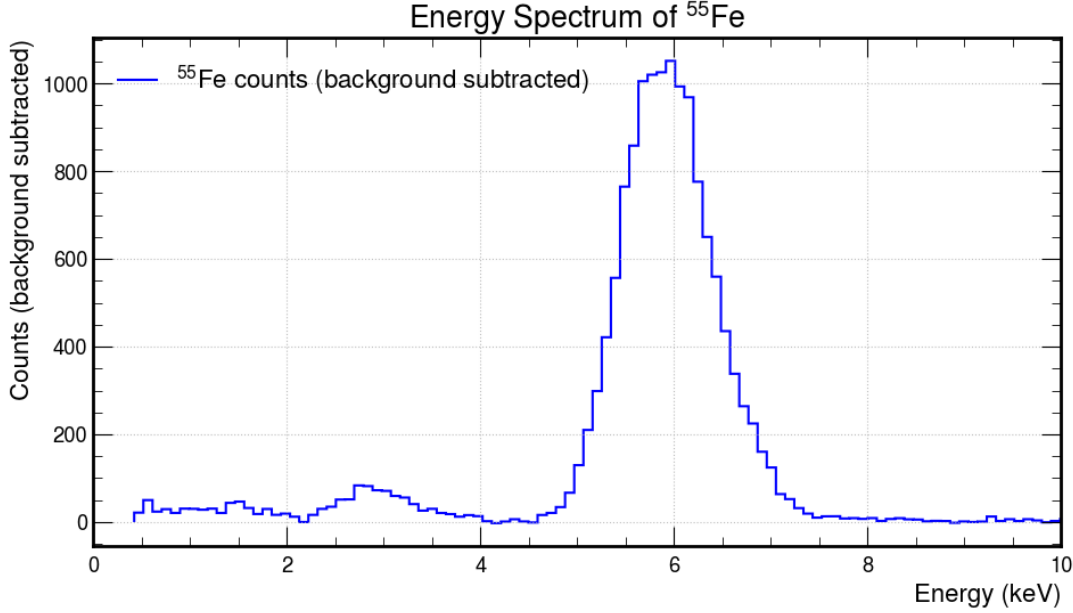


Figure 12: Measured energy spectrum for ^{55}Fe

4.4 Identifying peaks in the ^{55}Fe spectrum

The energy spectrum in the $E < 10$ keV regime measured for the ^{55}Fe sample is shown in figure 12. This regime was chosen so that features of interest are most prominent. The plot is obtained by applying the energy calibration fit to the raw data. Background radiation has been subtracted from the spectrum.

The major peaks in the ^{55}Fe spectrum were used for energy calibration. We verify that these align with the expected energies. We also make a double Gaussian fit for the escape peaks. These fits are shown together in figure 13 and individually in more detail in figures 14, and 15.

Figure 14 showcases the fit to the main photopeaks in better detail. We obtain the peak locations of (5.838 ± 0.018) keV and (6.491 ± 0.063) keV which both contain the true peaks (5.8592 keV and 6.49045 keV) in their uncertainties. This is to be expected, as these were used for calibration of the energy axis.

More interestingly, in figure 15 we identify the escape peaks. The first escape peak is located (2.979 ± 0.042) keV below the corresponding photopeak while the second escape peak is found to be (2.894 ± 0.182) keV below the main photopeak. These energies are in alignment with what one would expect for an escape peak using Argon (P-10 is 90% Argon) as the detector gas [8].

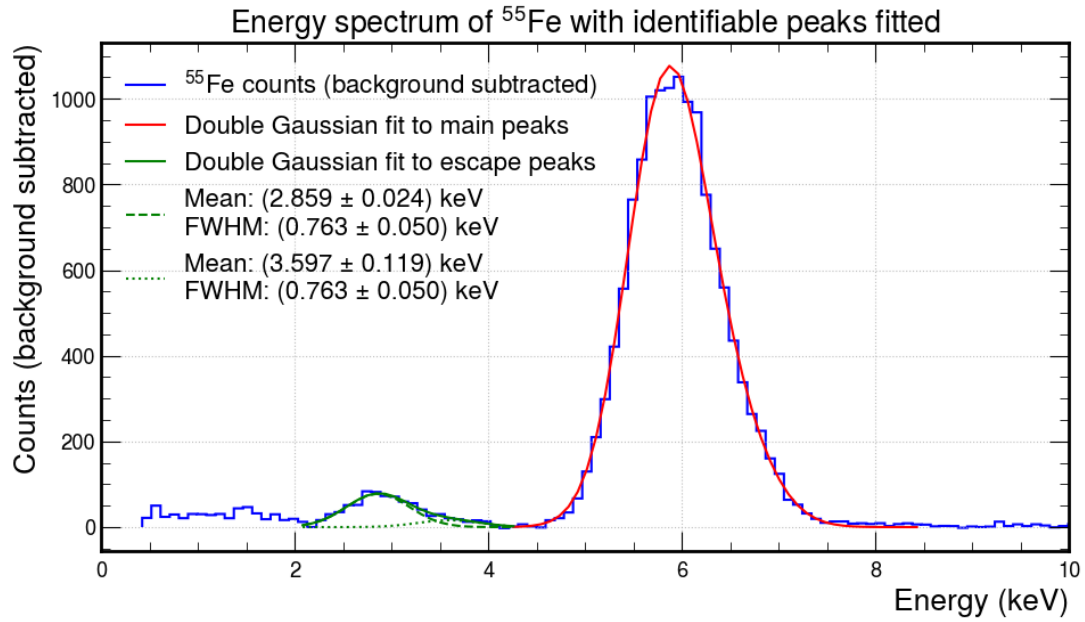


Figure 13: A fit to all identifiable peaks in the ^{55}Fe spectrum. We identify the two main photopeaks and their escape peaks.

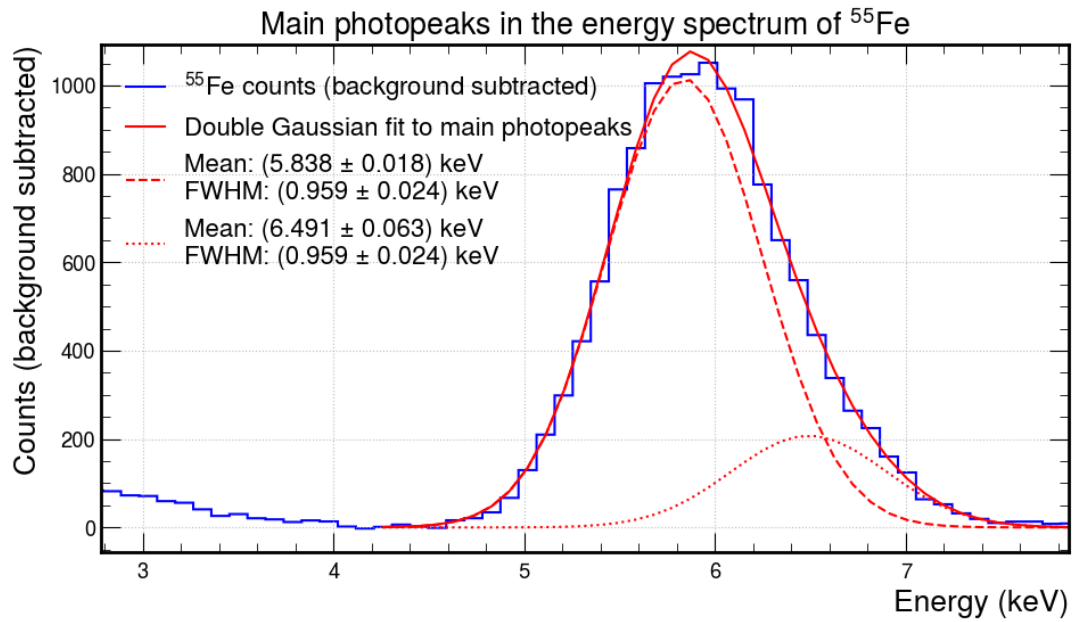


Figure 14: A fit to the two main photopeaks of ^{55}Fe . The peaks should represent the known 5.8592 keV and 6.49045 keV peaks. These are contained within the uncertainty.

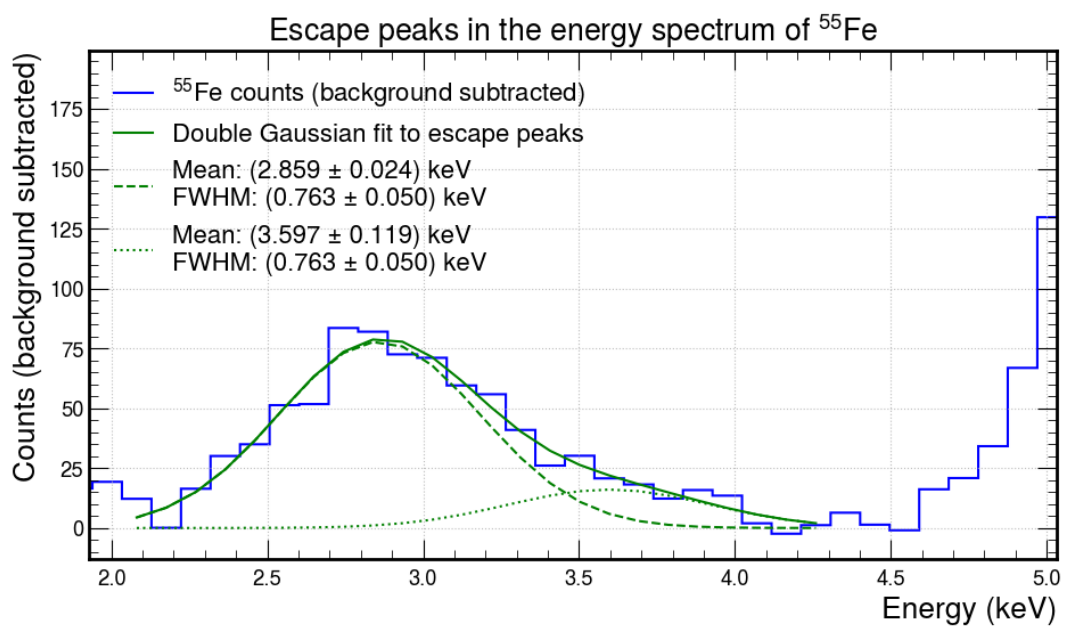


Figure 15: A fit to the double escape peak caused by the two primary photopeaks

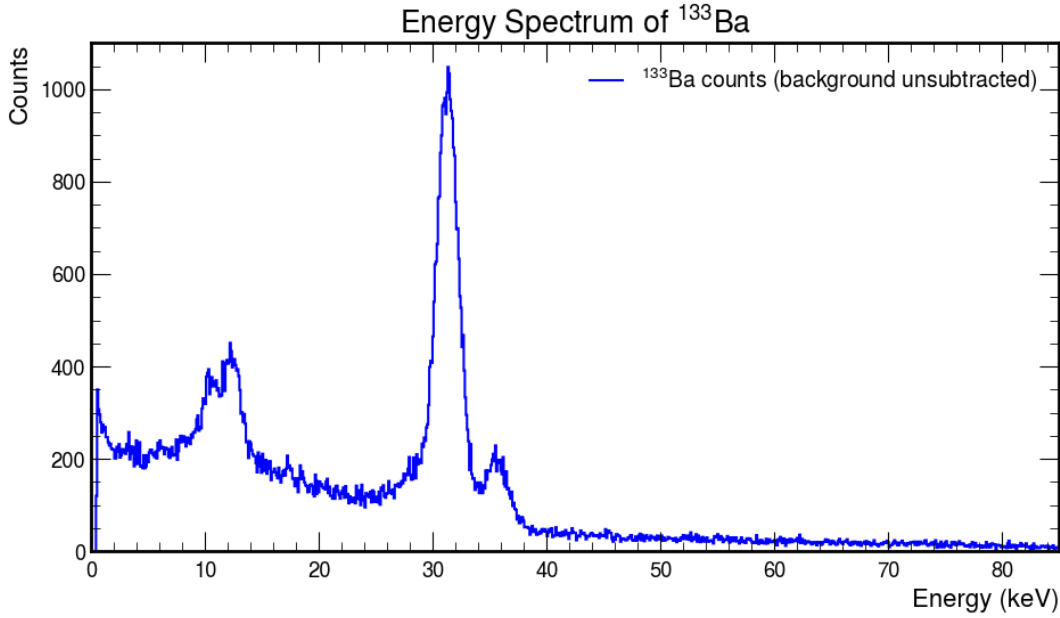


Figure 16: ^{133}Ba spectrum.

4.5 Identifying peaks in the ^{133}Ba spectrum

The ^{133}Ba spectrum was measured with a collimator in order to discern the higher energy peaks better. The collimator is made of lead and makes coincident events less likely, thus preventing signal overlap and improving the resolution. We do not perform background subtraction for this measurement, as they were made under different conditions to the background measurement. This is despite the fact that the background is unlikely to be significantly different for the collimated measurement, as it is largely made up of a muon background which could still pass through the lead collimator. We assume no significant complications to arise from this due to the relatively low background, especially in the $E > 10$ keV range. In figure 16 we plot the full measured energy spectrum for ^{133}Ba .

Literature suggests that ^{133}Ba has a L -line X-ray at 12.327 keV. We also expect two $K\alpha$ X-rays at 31.817 keV and 32.194 keV with intensities 15.1% and 27.6% respectively. Additionally, there exist three $K\beta$ X-rays at 36.304 keV (2.64%), 36.378 keV (5.10%) and 37.255 keV (1.61%). The reported energies vary, with other sources reporting similar peaks within 1 – 2 keV intervals.¹ At higher energies we also expect a peak at 53.1622 keV (2.14%), however due to the low intensity we are unlikely to observe this. A prominent peak is expected at 80.9979 keV (32.99%) [7].

Comparing the measured energy spectrum for the collimated ^{133}Ba source, to the literature values we are able to identify many of the same peaks. There are 4 visually identifiable peaks in figure 16. Gaussians have been fit to them in figure 17. We observe a peak around 12.2 keV, which coincides with the literature peak at 12.327 keV very nicely within the FWHM of the measured peak. Besides this peak, we also observe a peak around 10.0 keV,

¹See the two gamma and X-ray radiation lists in [7] for example.

which we speculate to be an escape peak of the L -line X-ray peak. There seems to be a rise toward the low energy regime, which could be explained by a reported peak at 4.47 keV [7], however this energy seems too high. The rise toward zero may also simply be an artefact of the muon background.

At higher energies, we have good agreement with the reported $K\alpha$ and $K\beta$ X-rays. We observe the $K\alpha$ peak at 31.228 keV, although literature suggests it to be slightly higher (around 32 keV). The literature reported value is within the FWHM of the peak. We are also confident in this being the $K\alpha$ peak due to the large intensity. We did not fit multiple Gaussians to the peak, as the fitting routine could not do this without very generous initial guesses. We also have good agreement with the $K\beta$ peak which we expect around 36.5 keV, but we measured around 35.5 keV. We do not resolve the individual peaks contributing to this peak, and therefore do not fit for them. With the variability in the literature reported values, the peaks are in the correct place. We also observe the expected lower intensity for the $K\beta$ peak.

Looking at figure 16, we do not see the prominent peak at ~ 80 keV. Similarly we do not see the low intensity peak around 53.2 keV. It is possible that we may have been able to resolve the 80 keV peak with a larger bias voltage.

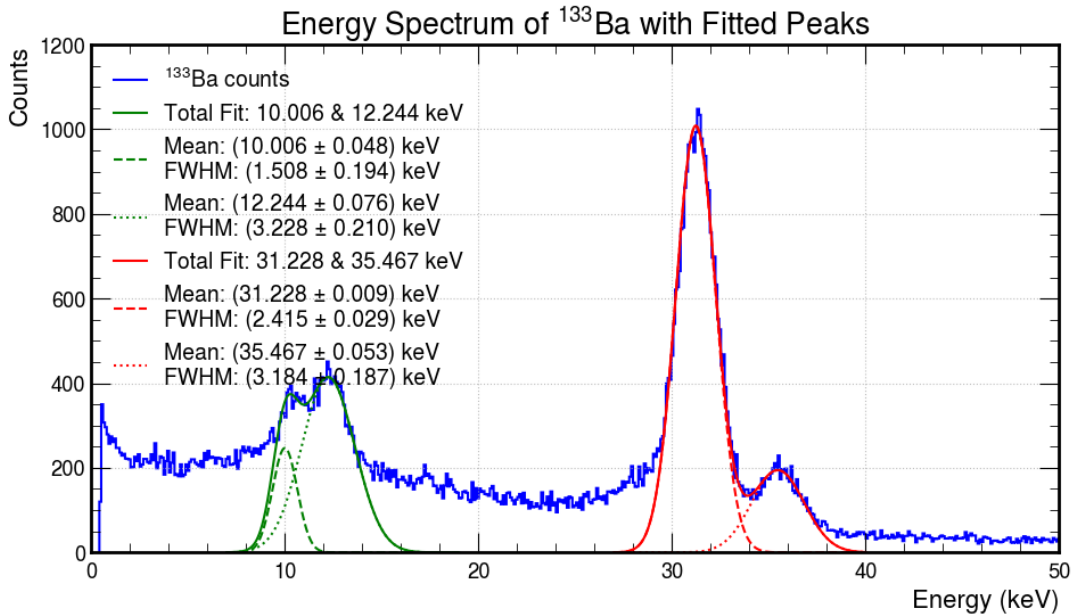


Figure 17: Peaks fitted to visually identifiable peaks in the ^{133}Ba spectrum.

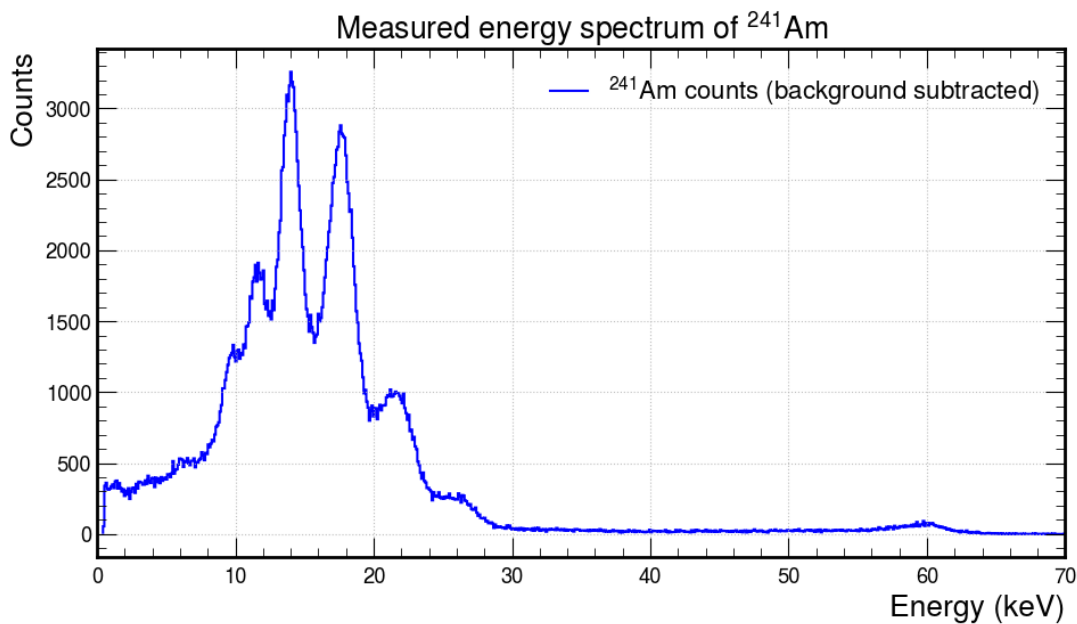


Figure 18: Full energy spectrum of ^{241}Am with background subtracted

4.6 Identifying peaks in the ^{241}Am spectrum

The full measured energy spectrum for the ^{241}Am measurement is shown in figure 18. We have already identified the rightmost peak around ~ 60 keV as it was used in the energy calibration. We will investigate the peaks present on the left-hand side in the $E < 30$ keV regime. By visual inspection there appear to be 4 peaks, and 3 bumps which may be peaks. Using the NuDat 3.0 nuclear database [7] we are able to confidently identify one low energy peak for Americium, namely at 13.9 keV with an intensity of 37%, which should also be present in our data. The database also suggests a peak present at 26.3446 keV with an intensity around 2.27%, although it is unlikely that this is contributing to the full peak which we observe around that value due to the relatively low intensity. A paper by D. Cohen, 1987 [3] suggests that L and M shell de-excitations of the ^{237}Ne atom which is the daughter nucleus for the α -decay of ^{241}Am , results in over 20 different peaks in the range between 2 – 25 keV. We include a good figure from ref. [5] to summarize the L -transitions which are observed in the ^{241}Am decay in figure 19.

Line	Transition	Observed energy (keV)	Siegbahn's extrapolated energy (keV)	Corrected relative intensity
Np $L_{\alpha 2}$	L_3-M_4	13.79 ± 0.04	13.76	7
Np $L_{\alpha 1}$	L_3-M_5	13.98 ± 0.04	13.94	64
Np L_{η}	L_2-M_1	15.90 ± 0.06	15.85	2
Np $L_{\beta 6}$	L_3-N_1	16.15 ± 0.06	16.11	1
Np $L_{\beta 2}$	L_3-N_5	16.89 ± 0.02	16.88	13
Np $L_{\beta 4}$	L_1-M_2	17.10 ± 0.02	17.06	16
Np $L_{\beta 5}$	$L_3-O_{4,5}$	17.58 ± 0.06	17.51	3
Np $L_{\beta 1}$	L_2-M_4	17.78 ± 0.01	17.74	100
Np $L_{\beta 3}$	L_1-M_3	18.02 ± 0.02	17.98	13
Np $L_{\gamma 5}$	L_2-N_1	20.15 ± 0.08	20.09	2
Np $L_{\gamma 1}$	L_2-N_4	20.82 ± 0.02	20.77	34
Np $L_{\gamma 2}$	L_1-N_2	21.12 ± 0.04	21.09	8
Np $L_{\gamma 3}$	L_1-N_3	21.31 ± 0.04	21.34	7
Np $L_{\gamma 6}$	L_2-O_4	21.51 ± 0.04	21.49	10
Np $L_{\gamma 4-\gamma 4}$	$L_1-O_{2,3}$	22.19 ± 0.04	22.19	3
Am $L_{\alpha 1}$	L_3-M_5	14.66 ± 0.03	14.61	10
Am $L_{\beta 1}$	L_2-M_4	18.76 ± 0.04	18.80	6

Figure 19: L x-rays observed in the decay of ^{241}Am (ref. [5]) .

In figure 20 we present fits to each of the candidate visible peaks. The full sum of the fitted peaks is presented in figure 21, this serves as a visual aid in gauging the goodness of fit obtained by the 7 gaussian functions.

The expected peak at 13.9 keV is present and can be fitted with a mean of (13.929 ± 0.012) keV. It is the peak with the greatest intensity, as expected.

The combination of X-rays around 16.15 – 18.02 keV have a relatively high intensity. These correspond rather well with the second most prominent peak which we observe around 17.485 keV. Therefore, this peak is likely a combination of all the $L\beta$ de-excitations of the Neptunium daughter.

The $L\gamma$ X-rays around 20.15 – 22.19 keV from [5] (figure 19) also correspond well with the measured peak around 21.4 keV. We thus identify this peak as a combination of the $L\gamma$ transitions in the de-excitation of the Neptunium daughter nucleus.

We do not obtain a peak at exactly 26.3446 keV but rather at (25.709 ± 0.222) keV. It is likely that the measured peak here is at least partly representative of the expected peak from literature.

We assume the others peaks to be a result of the unlisted lower energy peaks. Many of the peaks are also likely a sum of several escape peaks.

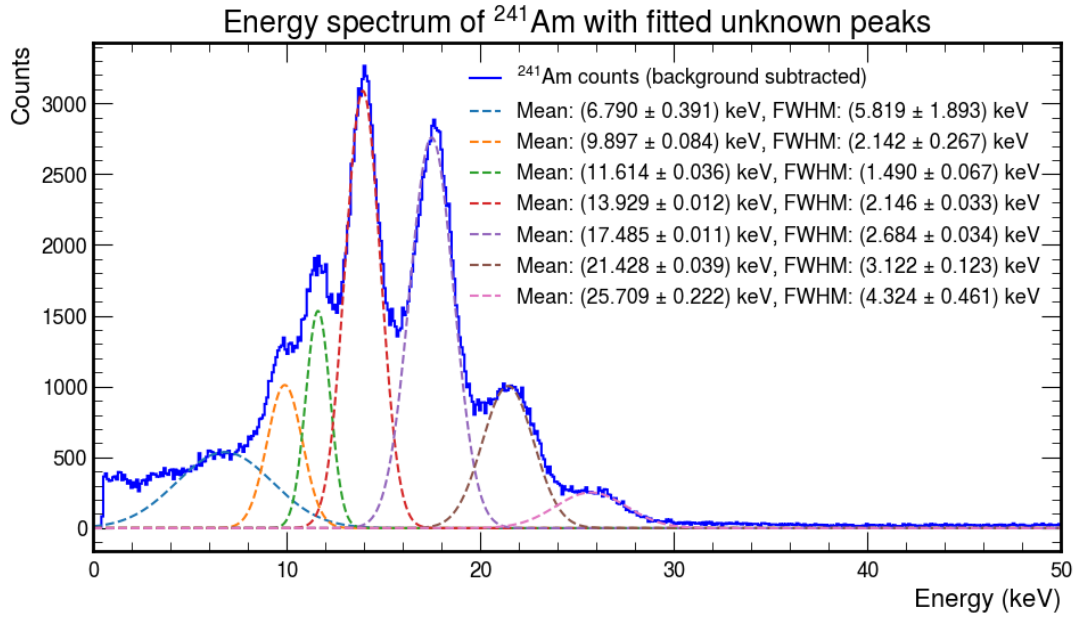


Figure 20: Gaussian fits to the 7 visually identifiable peaks in the ^{241}Am spectrum.

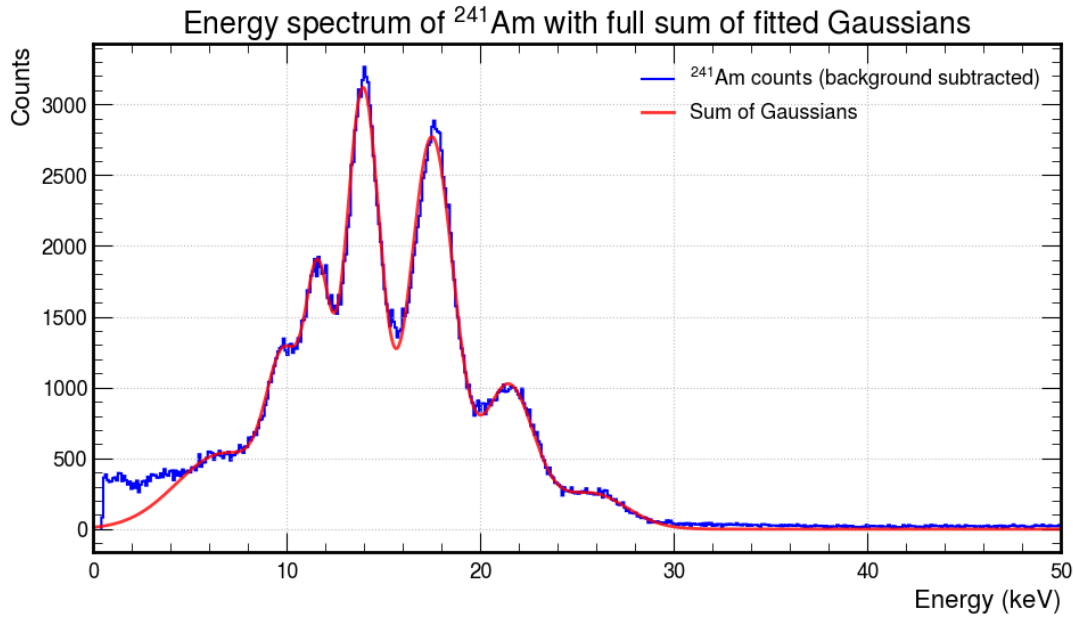


Figure 21: A sum of the gaussians presented in figure 20. This plot is intended to demonstrate the goodness of fit of all the fitted Gaussians together.

5 Conclusions

We constructed a proportional counter using an aluminium beverage can as a gas chamber as well as additional materials and tools provided by our lab supervisors.

The detector was operated using a P-10 gas mixture. After calibrating the detector and determining a linear relationship between pulse height and MCA channel output, we determined the multiplication factor of the gas multiplication in the detector at voltages between 900 V and 2200 V using the distinct peaks of Fe-55 and Am-241. We found that our detector performed as expected by theory [4] and had excellent resolution. With a fixed 1747V bias voltage we then took 10-30 minutes long spectral measurements of ^{55}Fe , ^{133}Ba and ^{241}Am sources. We were able to identify many known distinct, high-intensity peaks below 60 keV. However, our detector performed poorly above 60 keV, being unable to pick up the 80 keV peak of ^{133}Ba .

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Group Work Experience

We worked fairly well as a team. Eetu did most of the data analysis, while Jeff focused on theory and reporting the experimental setup. However, we could both consult the other when problems occurred in our respective tasks. Sharing the workload has decreased the burden of the work for both of us and allowed us to much more easily produce a high quality report than if only one of us had attempted it with the same time and resource constraints. It was of immense help that both of us were familiar with soldering, spectroscopy and data analysis and could thus both help with the more challenging parts of building the detector as well as identifying and categorising peaks. Most of the difficulties in the shared work arose from either conflicting schedules or differing stylistic visions for the final report document.